

scoreboard.vr

File location: \$VCS_HOME/doc/examples/nativetestbench/openvera/rvm/host_if/env/scoreboard.vr

```
/*
 *
 * Author:      Fabian Delguste
 * File:        $RCSfile: scoreboard.vr,v $
 * Revision:    $Revision: 1.5 $
 * Date:        $Date: 2003/11/14 14:45:32 $
 *
 ****
 *
 * Scoreboard module used to verify the
 * incoming and outgoing transactions correctness
 *
 ****
 */

#include "test_cfg.vrh"
#include "rvm_std_lib.vrh"

class scoreboard {
    rvm_log log;
    test_cfg cfg;
    // 32x8 register file
    bit[7:0] register_file[32];
    // Constructor
    task new(test_cfg cfg) {
        integer i;
        this.log = new("Scoreboard", "singleton");
        this.cfg = cfg;
        // initialize the register file with 0's
        for (i=0; i<32; i++) {
            register_file[i] = 0;
        }
    }
    // Methods allowing to update the 32x8 register file
    task write_register(bit [4:0] addr,
                       bit [7:0] data);
    task expect_register(bit [4:0] addr,
                        bit [7:0] data);
}
```

```

}

task scoreboard::write_register(bit [4:0] addr,
                               bit [7:0] data)
{
    if (addr > 7) {
        this.register_file[addr] = data;
    }
}

task scoreboard::expect_register(bit [4:0] addr,
                                bit [7:0] data)
{
    if (register_file[addr] !== data) {
        rvm_error(this.log, psprintf("Unexpected data read at 0x%02x: 0x%02x instead of 0x%02x",
                                     addr, register_file[addr], data));
        // Minimize the number of futur errors.
        register_file[addr] = data;
    }
}

```